Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor | Project Report

Abstract

This project examines shipment and sales records for Nassau Candy Distributor to summarize shipping activity by shipping mode, region, and city. A spreadsheet-based workflow was used to prepare the data, calculate shipping duration, create PivotTables, and build a dashboard with key performance indicators (KPIs). The report documents the approach and key descriptive findings. It is a project report, not a peer-reviewed or published research paper.

1. Introduction

Metric	Reported workbook value
Shipment records	10,194
Units	38,654
Sales	141,783.63
Gross profit	93,442.80
Average shipping days	1,320.84

Distribution operations depend on reliable movement of goods from factories to customers. Shipping performance affects customer experience, operating cost, and the ability to plan capacity. This analysis uses available order and shipment records to explore shipping patterns and identify areas that may warrant further investigation.

2. Objectives

The objectives are to: (1) organize and review shipment data; (2) calculate the interval between order and ship dates; (3) compare shipment counts and business measures across shipping modes and regions; (4) identify high-volume cities; and (5) present findings in a concise dashboard.

3. Data and Methodology

The workbook contains 10,194 shipment records. The analysis was completed in WPS Spreadsheet using a cleaned-data worksheet, calculated shipping days, PivotTables, and dashboard charts. Shipping days were calculated as Ship Date minus Order Date. PivotTables summarized records by Ship Mode, Region, City, and Region × Ship Mode. Dashboard KPIs included shipment records, units, sales, gross profit, and average shipping days.